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Data Channel	Single Side Band (SSB)
Modem	Modulation
Bit Error Rate (BER)	Digital Data
Mean-Bits-Between-Errors (MBBE)	

20. ABSTRACT *(Continue on reverse side if necessary and identify by block number)*

This test operations procedure (TOP) prescribes engineering test procedures necessary to determine the technical performance of special equipment required to condition digital data signals for transfer over military communication facilities; and the capability of these media to transmit the information accurately, relative to criteria expressed in the Required Operational Capability (ROC), Letter Requirements (LR), and Coordinated Test Program (CTP).

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Item 20 (cont)

The necessity for real-time exchange of digital data information among the dispersed elements of a military automatic data processing (ADP) system requires the use of all forms of communications media existing in the environment to provide the interconnecting paths. Since the characteristics of ADP signals and equipment, per se, are seldom wholly compatible with the characteristics of all types of transmission path, some form of interface equipment is required at the path terminals to attain compatibility.

An ADP-Communication interface may consist of a relatively simple device performing a single function or a complex aggregation of equipment performing several indirectly related functions. However, the basic function is to operate on digital data information in such a manner as to effect efficient transfer over conventional communication systems within the channel frequency spectrum available and the propagation constraints of the transmission media.

For the purpose of this document, "data transmission equipment" is defined as channel termination equipment employed for the sole purpose of data signal transformation. A selected communication channel terminated with this equipment is termed a "data channel". In other word, the role of a data channel is to convey information from a data "source" to a data "sink", without knowledge or manipulation of the information content of the data.

Various devices employed to accomplish the conversion process are generically termed "modems" (modulator - demodulator) and in the simplest form convert digital dc signals of the ADP equipments to/from digital ac (quasi-analog) signals for transmission over conventional circuits. A form of modem which performs the same basic function in telegraph/teletypewriter communications is exemplified in the long-time military standard TH-5 and TH-22 Telegraph Terminals. These devices convert the dc telegraph/teletypewriter signals to frequency-shifted audio tones for transmission over standard voice channels at information transfer rates up to 75 bits (binary digits) per second.

A further discussion of data transmission equipment is given in Appendix A of this TOP.

Since data transmission is a rapidly growing art, new equipment and systems will continue to evolve to meet data network requirements. Therefore, comprehensive engineering tests are necessary to determine technical performance adequacy with respect to specific requirements and existing equipment/ systems.

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US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

DRSTE-RP-702-105
Test Operations Procedure 6-2-065*
AD No. A134711

1 September 1983

DATA TRANSMISSION EQUIPMENT

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1.0 SCOPE. This test operations procedure (TOP) delineates general engineering test procedures for measuring and evaluating the technical performance and characteristics of data channel terminal equipment and communications system facilities with respect to their data transmission capability relative to criteria specified in the Required Operational (ROC), Letter Requirements (LR), and Coordinated Test Program (CTP).

This TOP excludes consideration of equipments which are designed or required solely to resolve automatic data processing (ADP) equipment anomalies or to perform ADP control functions external to a central processor or input output (I/O) device. Examples are:

a. Speed buffers or modulation rate converters

b. Error control systems

c. Code converters

d. Alphabet translators

e. Special security devices

To a certain extent, the tests contained herein are predicated on the use of special instrumentation comprising a data transmission test facility as described in paragraph 2.1. However, individual components can be employed with appropriate adaptation.

*This TOP supersedes MTP 6-2-065, 23 September 1969.

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1.1 Common Engineering Tests. Not included in this TOP are the following Common Engineering Tests which may apply to this commodity.

a. 1-2-610, Human Factors

b. 6-2-500, Physical Characteristics

c. 6-2-504, Maintenance/Maintainability

d. 6-2-508, Electromagnetic Vulnerability

e. 6-2-509, Electromagnetic Compatibility

f. 6-2-514, Electrical Power Requirements

g. 6-2-516, Adequacy of Shelter and Van Mounted Lighting, Ventilation, Air Conditioning, and Heating Equipment.

h. 6-2-520, Transportability of Communication, Surveillance, and Electronic Equipment.

i. MIL-STD-810D, Environmental Test Methods and Engineering Guidelines.

j. 6-2-507, Safety and Health Evaluation - Communications/Electronic Equipment.

2.0 FACILITIES AND INSTRUMENTATION

2.1 Facilities. Test shall be performed at a "data transmission test facility." This facility will consist of locally assembled instrumentation complexes (transmitting and receiving). Major functions are:

a. Conventional transmission (circuit quality) measurements of communication channels proposed for data transmission.

b. Determination of overall data channel quality by measurement and analysis of actual data transmission, i.e., rate, characteristics, and distribution of errors in received test transmission.

In a typical test operation involving, for example, a point-to-point communication link, the test channel characteristics are first measured and recorded. The transmitting unit is then set up to send data test signals in a definite pattern at a prescribed rate. The receiving unit operates on the received test pattern in a number of ways to determine the extent and characteristics of any degradation imposed by the test channel. Correlation of data, events and processing operations is in real-time.

2.2 Instrumentation and Equipment. Instrumentation and equipment at the transmitting unit shall include standard test equipment items for (1) straightaway transmission, (e.g., signals and noise generators, frequency counters, oscilloscope, delay test set) and, (2) a digital signal/pattern

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generator with modems (standard or test item) for generating data test signals. Instrumentation and equipment at the receiving unit shall include, but not be limited to (1) test equipment for the basic channel measurements, (e.g., level meter, distortion analyzer, delay test set) and, (2) compatible modems, real-time clock, digital pattern generator (identical to transmitter), error count subsystem, computer subsystem, and recording subsystem for analysis of received data signals. An RF field strength measuring system is required additionally for correlation of radio path vagaries with received data test signals, although multi-hop systems introduce many unknown factors.

A multi-channel magnetic tape recorder/reproducer is employed to simultaneously record the received test pattern, test events, RF signal, and real-time code with provision for rapid search for and play-back of specific recorded test intervals.

2.3 Characteristics/Requirements. The characteristics and accuracies of the above test instrumentation are determined by the performance specifications of the individual item to be tested and the circuits in which it is to be used. Select test instrumentation having an accuracy of sufficiently higher quality (10:1) than that of the test item. Calibration of all test instrumentation shall be traceable to the National Bureau of Standards. The above listed major facilities, instrumentation and equipment will provide the necessary characteristics and set-up required to perform the subtests outlined by this TOP.

3.0 PREPARATION FOR TEST

a. Review TECOM Pamphlet 70-3, Research, Development, and Acquisition and Project Engineers' Handbook for guidance on test planning, execution, reporting, and post-test activities.

b. Establish and/or continually maintain a readily accessible project log and project file.

c. Review local installation project office handbook, standing operating procedures (SOP) and implementing directives which govern the administrative processes of preparing test plans, conducting tests, preparing reports, reporting to the Test Resources Management System (TRMS), and budgeting.

d. Acquire and review all descriptive, instructional, and specification material on the test items issued by Government and contractor(s) against the test plan's subtest objectives, criteria, facility(ies), and instrumentation requirements.

e. Determine the scheduled availability of the test item.

f. Insure availability of appropriate facilities, and coordinate the test support requirements including personnel, equipment, maintenance, spare parts, and instrumentation.

g. Review the detailed test plan.

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h. Record, as a minimum, the following data:

(1) Nomenclature, serial number(s), manufacturer's name, and function of the item(s) under test.

(2) Nomenclature, serial number, accuracy tolerance, calibration requirements, and last calibration date of test equipment selected for the tests.

(3) Damage to the test item(s) incurred during transit and/or manufacturing defects.

(4) Test item photographs.

i. Establish instrumentation or measurement system mean error and standard deviation of error.

j. Determine test item sample size.

k. Insure that arrangements for supporting and participating agencies, activities and facilities have been made, that authorization for electromagnetic radiation at specific frequencies, power levels and modulations for required periods has been obtained, that all personnel have been briefed on the purpose of the tests and the results expected, and that appropriate security measures are instituted, as required, to safeguard classified material and data.

l. Prior to beginning any subtest, verify correct power source, necessary test instrumentation and interconnecting cabling, and that the equipment is aligned, if necessary, as specified in the pertinent operating instructions. Insure, insofar as possible, that system being tested represents an average equipment in normal operating condition.

4.0 TEST CONTROLS

a. Organize test team and establish responsibilities for test conduct, reporting, and data control.

b. Select test equipment having an accuracy at least 10 times greater than that of the function to be measured.

c. Familiarize all test personnel with the test item instructional material.

d. Prepare adequate safety precautions to provide safety for personnel and equipment and insure that all safety SOPs are observed throughout the test.

e. Inspect the test item thoroughly for obvious physical and electrical defects. Note and correct all defects before proceeding with testing.

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f. Prepare, disseminate, and continually review record forms for systematic entry of data, chronology of tests, and analysis in final evaluation of the test item.

g. Prepare and implement a test item sample plan to insure that enough samples of all measurements are taken to provide statistical confidence of final data in accordance with TOP 3-1-002 (Confidence Intervals and Sample Size).

h. Perform such preliminary tests as necessary to assure the test item(s) is (are) testable.

5.0 PERFORMANCE TESTS

NOTE: Modification of these procedures shall be made as required by technical design of the test item and availability of test equipment, but only to the extent that such modified procedures will not affect the validity of test results.

5.1 Test Methods. Divide the specific tests to be performed into component (laboratory-type) and system (field) tests. These tests and their intended objectives are as follows:

a. Component Tests - The objective of these tests is to determine the input-output and performance characteristics of a type data modem operating as the test item in a simulated data link.

b. Systems Tests - The objective of these tests is to measure the performance of the test item with respect to the test system's inherent characteristics involving digital data traffic.

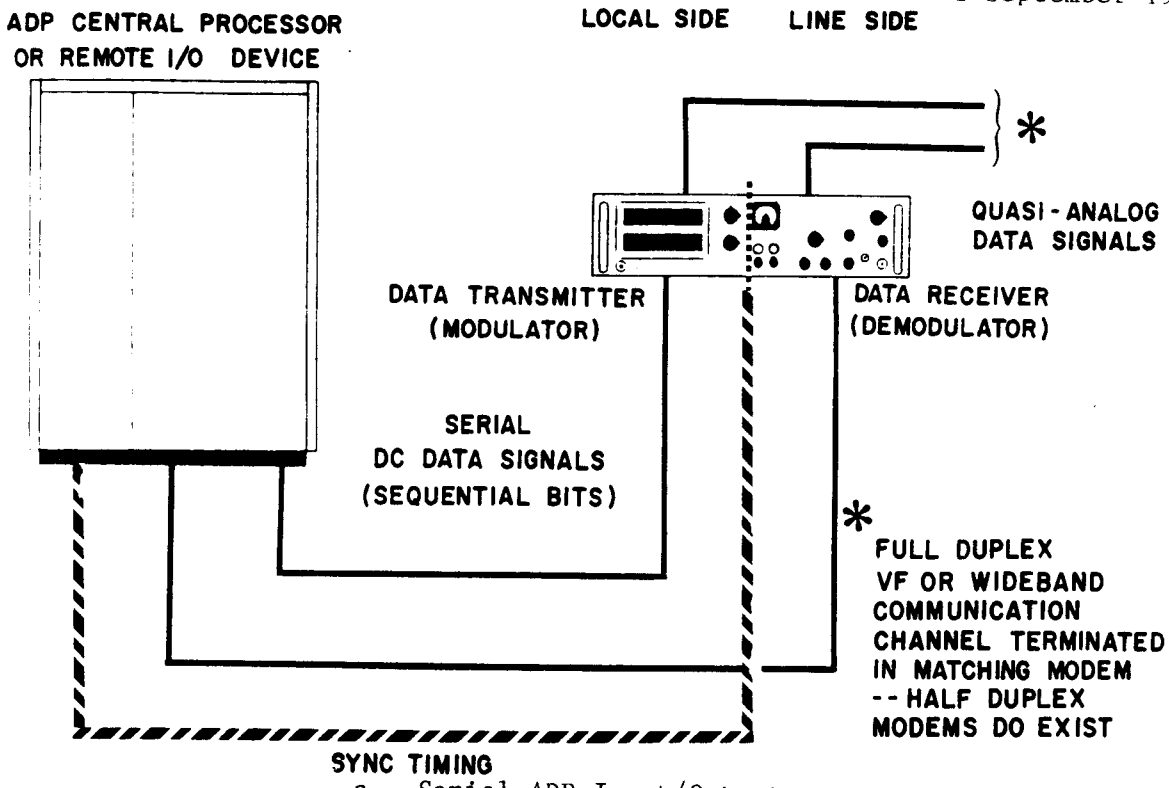
The closed-system and open-field tests are essentially identical. The closed-system tests are required as necessary to determine the terminal equipment performance parameters from which the open-field tests determine the effect of the overall real-system variables.

5.1.1 Common Tests

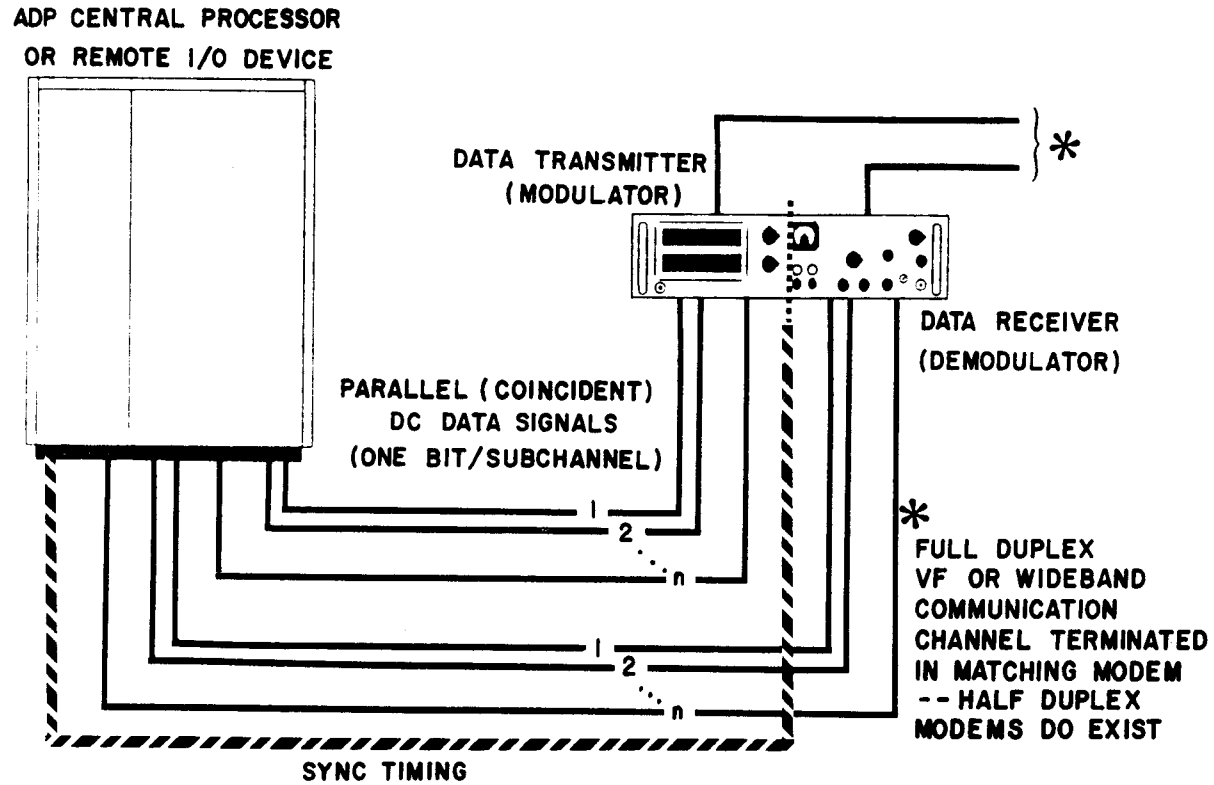
NOTES: 1. These tests encompass the basic characteristics generally common to data modem equipment and shall be conducted in a laboratory-type environment. A data transmission test facility may be utilized with the back to back configuration over a simulated line to provide the setup shown in figure 2.

2. The inputs and outputs of generic data modems (serially and parallel) are illustrated in figure 1.

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a. Serial ADP Input/Output.



b. Parallel ADP Input/Output.

Figure 1. Basic characteristics of type data modems.

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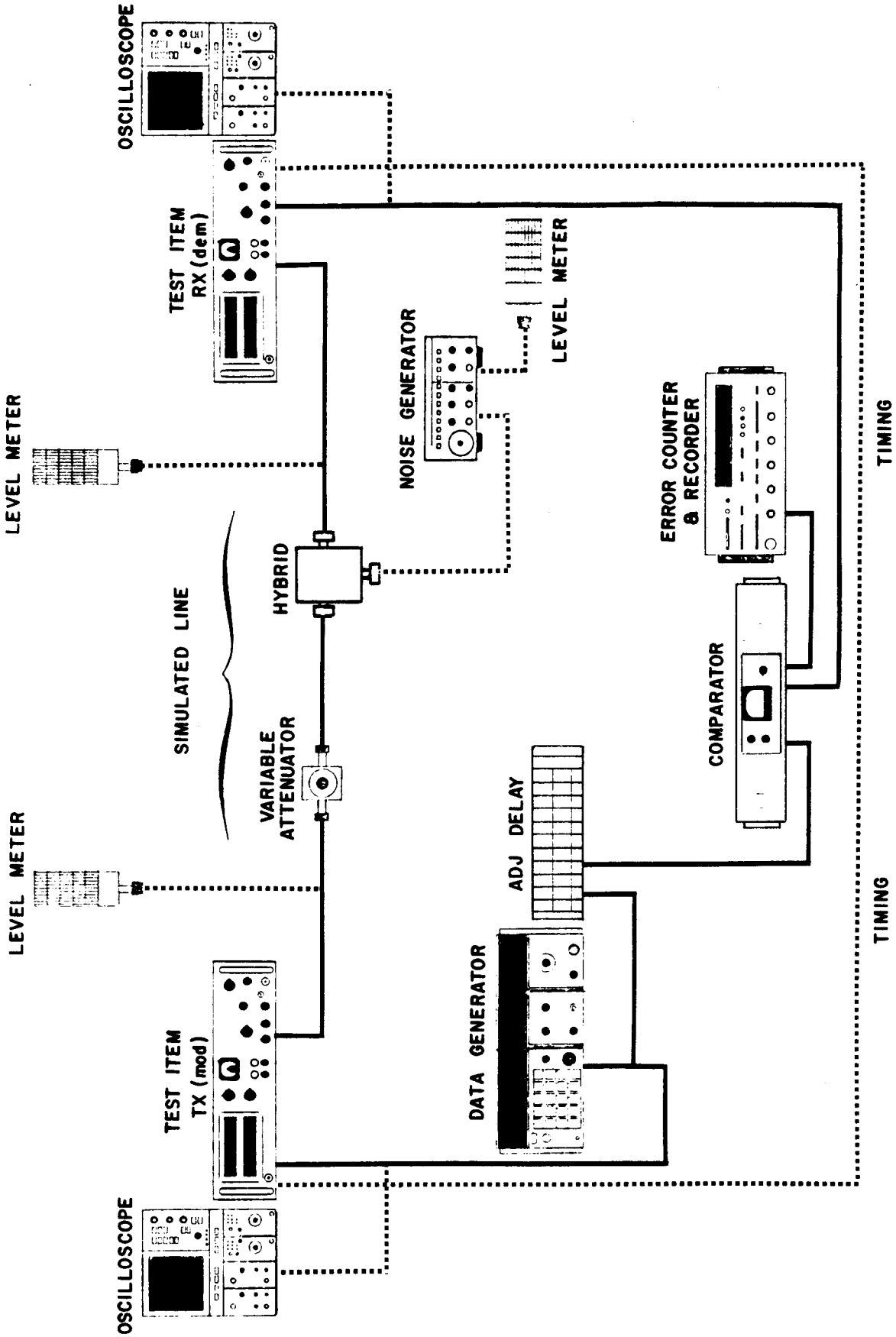


Figure 2. Generalized component test setup.

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5.1.1.1 Terminal Impedance. Prior to setting up the test items as shown in figure 2, perform the following measurements on the test items:

a. Measure the output and input impedance at selected test frequencies representative of the design line frequency band.

b. Measure the output impedance of the timing source at test frequencies commensurate with the test item bit rates.

c. Measure the capacitance and dc resistance of the dc data input(s) and output(s) (local side).

5.1.1.2 Operating Parameters

NOTES: 1. These tests shall be performed with the test items set up and operating as a simulated data link as shown in figure 2.

2. Data test signals shall be transmitted in blocks of established length (total bits) for each test for subsequent determination of error rate.

5.1.1.2.1 Timing. Determine timing signal characteristics as follows:

a. Measure the frequency and level of the reference timing output signal and check for stability over a minimal 14-hour period. Photograph an oscilloscope presentation of the waveform, or use a programable oscilloscope that produces hardcopy data.

b. Repeat step a above at each bit rate for which the test item is designed.

5.1.1.2.2 Transmitter Output Characteristics. Determine transmitter output characteristics as follows:

a. Measure the transmitter output power level (dBm) over the range of the output control (minimum, maximum, and mid-range).

b. Repeat step a above at each bit rate for which the test item is designed.

NOTE: A programed power reading meter with hardcopy readout may be used.

5.1.1.2.3 Line Signal Frequency Spectrum. Determine the line signal freqency spectrum characteristics as follows:

a. Determine the line signal frequency spectrum (output level at nominal mid-range) by scanning the line signal in frequency with a wave analyzer at the transmitter output, using maximum wave analyzer selectivity (minimum bandwidth). Measure amplitude at sufficient points between the lower and upper minimum measurable amplitude points to adequately describe the spectrum, or use a programable spectrum analyzer that converts the CRT readout to a plotter.

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b. Repeat step a at each bit rate for which the test item is designed.

5.1.1.2.4 Receiver Characteristics. Determine receiver characteristics as follows:

a. Measure the receiver sensitivity and dynamic range by varying the line signal level as measured at the receiver input, in combination with the receiver gain control (if provided) to determine the operating range over which the receiver correctly assumes the intended binary state.

b. Photograph oscilloscope presentations of the receiver output waveforms at selected points during step a above to depict the extent of output signal degradation with respect to a varying input.

c. Repeat steps a and b at each bit rate for which the test item is designed.

5.1.1.3 Noise Tolerance

NOTES: 1. This test shall be performed with the test items set up and operating as a simulated data link as shown in figure 2.

2. The data signal generator shall be programed for a repetitive uniform bit pattern and operated at selected bit rates and a nominal output level.

3. Data test signals shall be transmitted in blocks of established length for each test.

a. Conduct a data transmission test without external noise to establish a reference with a constant nominal received signal level, e.g., -10 dBm.

b. Repeat step a above with random noise added in increments to produce signal-plus-noise-to-noise (S/N) ratios in steps ranging, for example, from 40 dB to 5 dB.

NOTE: For the random noise, a random (white) noise generator shall be used which has a flat power density spectrum equal to or greater than the frequency band of interest as determined in paragraph 5.1.1.2.3a.

c. Photograph oscilloscope presentations of the receiver output (dc) signal at S/N ratios selected to correlate signal degradation with increasing error rate.

d. Repeat steps b and c at each bit rate for which the test item is designed.

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e. Repeat steps b, c, and d, utilizing an impulse noise generator which has output characteristics (pulse duration, repetition rate, polarity) commensurate with the data test signals or specific test requirements; impulse noise amplitude shall be the effective (rms) value measured by the level meter in dBm.

5.1.2 System Tests

NOTES: 1. Systems tests are based essentially on two correlated subtests, i.e., measurement of channel electrical characteristics and measurement of data transmission in terms of the error characteristics. For test description purposes, the simplified system of figure 3 represents the essential elements-terminals/stations, paths, and derived data channels of typical systems. In this con text, nominal 4-kHz analog channels are assumed; data modems are included in the Data Transmission Test Facility (DTF).

2. Reference is made to MTP 6-2-200, PCM Multiplexers for methods of circuit quality measurement, particularly as applied to multi-channel systems.

3. All test data recordings shall include real time in code in order to effect precise correlation of events in subsequent analyses.

5.1.2.1 Closed-System (Back-to-Back) Tests

a. Connect the system terminals/stations back-to-back as represented by the simulated path of figure 3.

b. Perform the following standard transmission (circuit quality) measurements on the data channel(s) with noise source inoperative and the signal strength at a minimum to avoid overloading the receive terminals:

(1) Frequency response (loss versus frequency)

(2) Envelope delay distortions

(3) Total harmonic distortions

(4) Circuit noise

(5) Crosstalk

c. With the equipment set up as in step a above, adjust and maintain input and output data channel signal levels to the nominal system operating levels (output level = test signal plus circuit noise), and conduct data transmission tests by transmitting data test signals.

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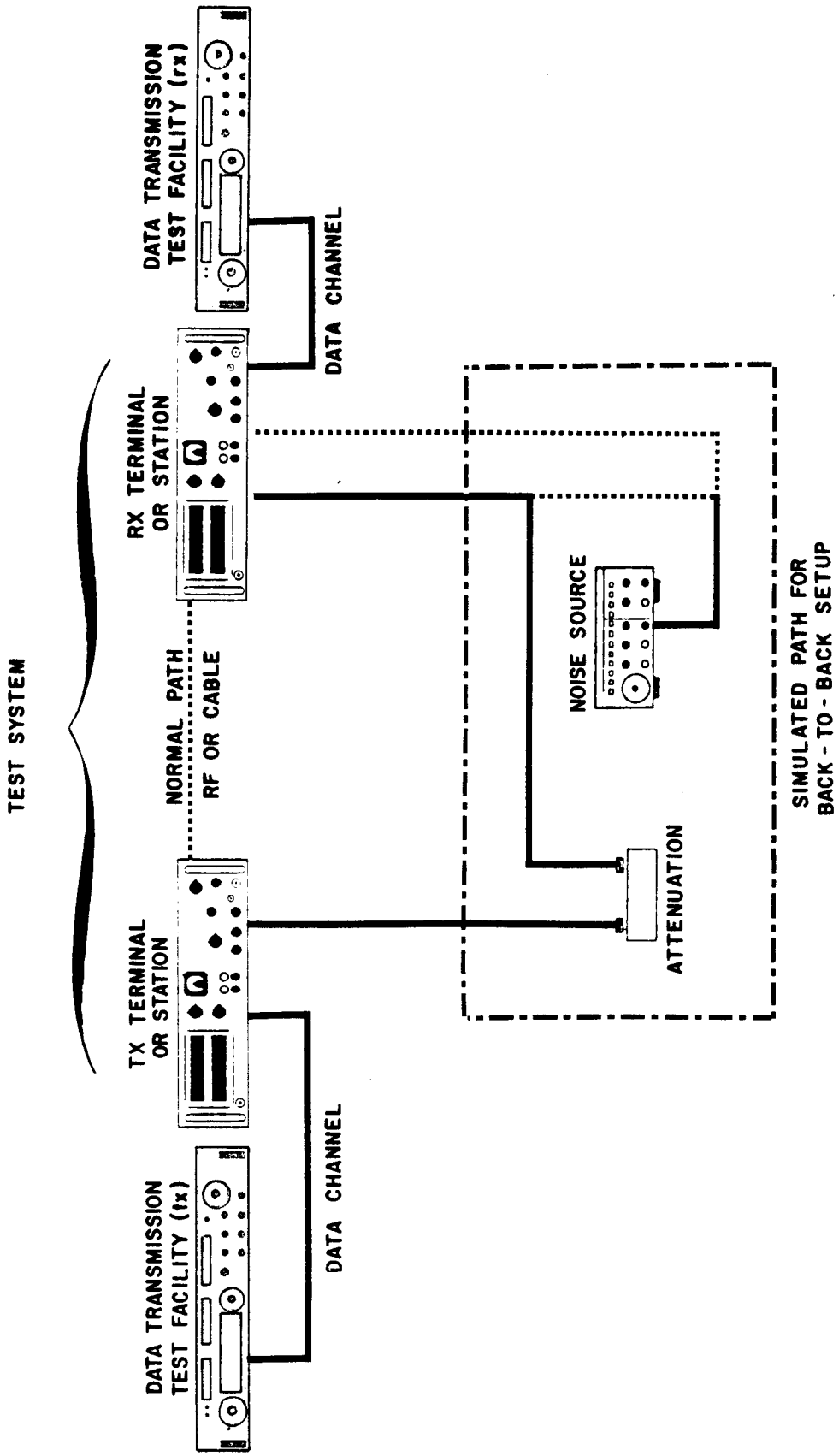


Figure 3. Simplified system test setup.

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NOTES: 1. Transmission shall be made in "test blocks" to test lengths in accordance with appendix A.

2. Tests shall be conducted at each bit rate indicated by the type of test item and/or test directive.

d. During the performance of step c above, re-run circuit quality measurements (step b) as often as necessary to detect variations and when investigating unacceptable test results (app A).

e. Following the test run at each bit rate, compute and record the signal-plus-noise-to-noise ratio of the receive channel levels measured in step c above.

f. Repeat steps c and d for noise tests described in paragraph 5.1.1.3 modified as follows:

(1) Introduce the random and impulse noise in turn at a point appropriate to the test system (fig. 3).

(2) Adjust the noise level in discrete steps to produce receive S/N ratios in decrements (e.g., 5 dB) from the value(s) determined in step e above to the S/N value at test truncation (app A). Completely automated procedures can save time, money, and solve accuracy problems.

5.1.2.2 Open-Field Tests

a. Install tactical communication system/facilities that are designated as test items for data transmission tests in a field environment with terminals/stations deployed at realistic system ranges selected to provide maximum fade margin and near maximum S/N ratio. General test setup shall be as shown in figure 2 (normal path).

NOTE: Due to the widely differing system characteristics and possible configurations, test setups and preparations shall be dictated by the type of system and available support facilities.

b. Insure, insofar as practicable, that system test configurations include ancillary facilities such as remote operation equipment, path, panels, and terminal wire and cable.

NOTE: Data modems required to terminate system test channels, but which in themselves are not test items, shall be selected on the basis of compatibility with the test channels and compliance with test requirements. In the event a communication system/facility and a specific data modem are both designated as test items of an integrated test, test procedures and results shall be differentiated as required.

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c. Conduct circuit quality and data transmission tests as outlined in paragraph 5.1.2.1 with applicable modifications as follows:

(1) Omit the stepped S/N tests of paragraph 5.1.1.3 using external noise sources.

(2) All radio system tests shall include a continuous RF signal strength measurement and recording at the received site, correlated in real time with other test events.

(3) Tests of systems employing SSB modulation, i.e., radio or frequency-division-multiplex (FDM) systems, shall include a channel frequency shift (frequency translation) measurement over the band of interest.

(4) Systems in which one terminal is normally in motion during real-test transmission shall be tested in typical configurations to determine the effect of the Doppler shift on data transmission accuracy. Airground and tactical vehicular communication systems are in this category. Controlled test conditions shall include variation of range, speed, and direction of travel of the moving terminal with respect to the fixed terminal during the data transmissions.

5.2 Data Required. Data to be recorded in addition to specific instructions listed below for each subtest shall include:

a. A block diagram of the test setup in each specified test. The block diagram shall identify, by model and serial number, all test equipment and interconnections (cable lengths, connectors, attenuators, etc.) and indicate control and dial settings where necessary.

b. Photographs or motion pictures (black and white or color), sketches, charts, graphs, or other pictorial or graphic presentation which will support test results or conclusions. If the cost is justified, video cameras and recorders could be used.

c. An engineering logbook containing, in chronological order, pertinent remarks and obserations which would aid in a subsequent analysis of the test data. This information may consist of temperatures, humidity, pressures, and other appropriate environmental data, or other descriptions of equipment or components, functions, and deficiencies as well as theoretical estimations, mathematical calculations, test conditions, intermittent or catastrophic failures, test parameters, etc., that were obtained during the test.

d. Test item sample size (number of measurement repetition).

e. Instrumentation or measurement system mean error stated accuracy.

5.2.1 Component Tests

5.2.1.1 Terminal Impedance. Record the following:

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a. Line input and output channel impedance (Z/0 or $R+_jX$) and the measurement frequencies.

b. Timing output channel impedance (Z/0 or $R+_jX$) and the measurement frequencies.

c. DC resistance (ohms) and capacitance (pf) of each dc data input and output channel or subchannel.

5.2.1.2 Operating Parameters

5.2.1.2.1 Timing. Record timing signal frequency (Hz) and level (dBm) at each design bit rate at intervals throughout the selected period. Include waveform photographs.

5.2.1.2.2 Transmitter Output Characteristics. Record the transmitter output level (dBm) and error rate information (number of bits in error/total bits transmitted) at each control setting and the composite data at each bit rate.

5.2.1.2.3 Line Signal Frequency Spectrum. Record line signal frequency spectrum data to include:

a. Transmitter composite output signal level (dBm).

b. Frequency (Hz) and signal level (dBm) at each selected frequency between minimum measurable amplitude points.

c. Wave analyzer selectivity (bandwidth) employed.

5.2.1.2.4 Receiver Characteristics. Record the following:

a. Received line signal level (dBm) at the selected points from minimum to maximum range.

b. Receiver gain control setting (if used) at each measuring point.

c. Error rate information obtained at each measuring point.

d. Waveform photographs correlated with the test data of a, b, and c above.

5.2.1.3 Noise Tolerance. Record the following for the noise tests (random and impulse) conducted at each bit rate:

a. Received line signal level (dBm).

b. Noise generator output levels (dBm).

c. S/N ratio (dB).

d. Error rate information at each S/N.

e. Waveform photographs.

f. Noise signal characteristics.

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5.2.2 System Tests

5.2.2.1 Closed System (Back-to-Back) Tests. Record the following closed system (back-to-back) test data:

a. Circuit quality measurements to include:

(1) Frequency response data in tabular form consisting of the received test tone level (dBm) measured at each test frequency.

(2) Envelope delay test data in tabular form showing the delay, in microseconds, measured at the incremental test frequencies.

(3) Total harmonic distortion test data in tabular form showing, for each test frequency, the measured signal + distortion and distortion only levels, both in dBm.

(4) Circuit noise level in dBm. Test data must include the line weighting and specific type of noise measuring set employed.

(5) Crosstalk test data consisting of the signal + crosstalk and the crosstalk only levels (dBm) measured in each selected channel combination.

b. Data transmission test data to include the following at each test bit rate:

(1) Test transmission rate (bits/second).

(2) Test length in terms of time and total bits transmitted.

(3) Bit-by-bit chronological error record.

(4) Correlative circuit quality deviations.

(5) Data S/N ratio with no external noise.

(6) Decremental data S/N ratios - random noise.

(7) Decremental data S/N ratios - impulse noise.

(8) Error rate at each data S/N ratio, (5), (6), and (7) above.

5.2.2.2 Open Field Tests. Record the following for the open-field system tests:

a. All items required in paragraph 5.2.2.1 (except the decremental S/N noise test).

b. Chart recording of RF signal strength versus real time (if applicable).

c. Profile diagrams of RF paths (as applicable).

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d. Meteorological conditions (as applicable).

e. Channel frequency translation (± Hz) at the selected test frequencies (SSB systems), as applicable.

f. Doppler shift test data (as applicable) to include detailed information on the terminal positions and direction and rate of travel correlated precisely with the data transmission test data.

6.0 DATA REDUCTION AND PRESENTATION

Processing of raw test data shall, in general, consist of organizing, marking for identification and correlation, and grouping the test data according to subtest title. Test criteria or test item specifications, where required, shall be noted on the test data presentation to facilitate analysis and comparison. Where necessary, test data measurements shall be converted to be compatible with units given by test criteria or specifications.

Primary test data shall be reduced by the most advantageous means and methods available; for example, the computer section of a data transmissison test facility may be configured and programed to provide some on-line computations and also be used for post-test-run data processing.

Specific instructions for the reduction and presentation of individual subtest data are outlined in the succeeding paragraphs.

6.1 Component Tests

a. Impedance, timing, and level range test data shall be presented in tabular form accompanied by annotated waveform photographs, or programable spectrum analyzers.

b. Line frequency spectrum data shall be presented as a graph of data line signal amplitude versus frequency. This can also be done with a programable spectrum analyzer.

c. Noise tolerance data shall be presented as a graph of error rate versus S/N ratio for each bit rate.

6.2 System Tests

a. Test data resulting from the closed-system and open-field tests may be presented separately or where the subtests in each configuration are comparable the two sets of test data may be combined in one graph and/or table to point up the effects of the real transmission path on data transmission quality.

b. Test data resulting from a series of circuit quality measurements made during one complete test shall be reduced to one graph/table for each channel parameter showing the mean values, the maximum deviations from mean, and the number of observations. In the event variation of any one parameter value has caused a degradation of data transmission quality, this data shall be presented separately, e.g., error rate variation versus the parameter variation.

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c. Data transmission test data shall be reduced to provide the following information appropriate to each test or subtest.

(1) Total bits transmitted.

(2) Total bits in error.

(3) Error rate = $\frac{\text{total bits in error}}{\text{total bits transmitted}}$

(4) Mean error free interval (bits) = $\frac{\text{total bits transmitted}}{\text{total bits in error}}$

(5) Mean burst length (bits).

(6) Total number of bursts.

(7) Mean interval between bursts (bits or time).

In addition to a statement of the above test results, data presentation may be made in a number of ways dependent upon test requirements and nature of the test and test item with the principal objective of depicting the relationship between any given data parameter and the test item characteristic or fortuitous variables. For example: error rate versus transmission (bit) rate; interval between bursts versus RF signal conditions; correlation of impulse noise and single-bit or bust errors; error rate versus S/N ratio.

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APPENDIX A

RATIONALE FOR DETERMINATION OF TEST PERIODS AND APPLICABILITY

Establishment of data transmission test and subtest periods having a certain flexibility with respect to duration and applicability is essential to insure adequate test results within practical limits and to preclude excessive testing time. Determination of these factors is influenced primarily by the test item type and concept of employment and the specified accuracy criteria.

Described briefly, the received test pattern is compared bit-by-bit with the transmitted test pattern generated within the receiving unit. A determination is made that each received bit is either correct, incorrect as a "1" to "0" transition, or incorrect as a "0" to "1" transition. The information is displayed, processed, and stored in a manner to provide a complete record of all correct and incorrect bits in the sequence of a record maintained of the total error-free one-second intervals. The test instrumentation is further configured, programed and employed to analyze the test data either during or subsequent to a test run and to provide information in printed or punched tape form, typical of the following:

a. Cumulative total of bits transmitted.

b. Cumulative total of bits in error.

c. Cumulative total of error bursts (burst: two or more consecutive bits in error).

d. Cumulative total of correct bits received.

e. Maximum, minimum, and mean interval (bits) between bursts.

f. Error rate $\frac{\text{(total bits in error)}}{\text{(total bits transmitted)}}$.

g. Total 1-second intervals.

h. Percent of error-free 1-second intervals.

The procedure described herein determines overall test length by progressively measuring test results against a specified criteria. The procedure is based on a "constant confidence level sequential test" for an exponential failure distribution (Ref 35). The procedure defines accept, reject, and continue testing regions for a given criterion of mean-bits-between-errors (MBBE) on a graph of total accumulated number of bits versus number of errors.

These regions are:

1. Accept if $N \geq 0.5\ N_o\ X^2(0.10;\ 2Y + 2)$

2. Reject if $N \leq 0.5\ N_o\ X^2(0.90;\ 2Y)$

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3. If $0.5\ N_o\ X^2(0.90;\ 2Y) < N < 0.5\ N_o\ X^2(0.10;\ 2Y + 2)$ continue testing unless $N > N$ truncation, at which time stop the test and calculate the lower limit 90-percent confidence interval on the true MBBE (u):

$$u \geq \frac{2N}{X^2(0.10;\ 2Y + 2)}$$

where

N = number of bits transmitted (total)

N_o = criterion MBBE

Y = number of errors (total)

$X^2(p;q)$ = the p-percentile of the chi-squared distribution for q degrees of freedom.

The three following methods and their associated graphs illustrate the use of the procedure for tests involving different MBBE criteria and test truncation points.

Method A - At the end of each test run, the accumulated total number of errors and the accumulated total number of bits transmitted will be checked against Table A-I or figure A-1. The test will be terminated if the intersection of the number of errors and bits transmitted falls within the "accept" or "reject" region. If the results fall within the "continue testing" region, testing will resume and the same checks will be made at the end of test run. However, if the results do not fall in either "accept" or "reject" by the time the total number of bits transmitted reaches 10^1 or 10^2 times the criterion, the test will be terminated. The observed MBBE will be reported and the lower limit of a 90% one-sided confidence interval will be calculated using Table A-I or the formula above.

TABLE A-I. DECISION TABLE FOR MBBE TEST TRUNCATION AT TEN TIMES CRITERION MBBE

Observed Errors	Reject if Total Bits Transmitted Are Below	Accept if Total Bits Transmitted Exceed	Truncation at 10 X Criterion Lower Limit of One-Sided 90% Confidence Interval
0	--	2.31 X Criterion	
1	0.11 X Criterion	3.89 X Criterion	
2	0.54 X Criterion	5.33 X Criterion	
3	1.11 X Criterion	6.69 X Criterion	
4	1.75 X Criterion	8.00 X Criterion	
5	2.44 X Criterion	9.28 X Criterion	
6	3.16 X Criterion		.949 X Criterion
7	3.40 X Criterion		.849 X Criterion
8	4.66 X Criterion		.769 X Criterion
9	5.44 X Criterion		.703 X Criterion
10	6.23 X Criterion		.649 X Criterion

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TABLE A-I. DECISION TABLE FOR MBBE TEST TRUNCATION AT TEN TIMES CRITERION MBBE (CONT)

Observed Errors	Reject if Total Bits Transmitted Are Below	Accept if Total Bits Transmitted Exceed	Truncation at 10 X Criterion Lower Limit of One-Sided 90% Confidence Interval
11	7.03 X Criterion		.602 X Criterion
12	7.83 X Criteron		.581 X Criterion
13	8.65 X Criterion		.527 X Criterion
14	9.47 X Criterion		.496 X Criterion

Method B - This method is the same as method A except that the MBBE criterion is 1 x 10^5 bits and the maximum test length is 1 x 10^6 bits. The curves in figure A-2 will be used.

Method C - This method is the same as method A except that the MBBE criterion is 1 x 10^5 bits. The maximum test length remains at 1 x 10^7 bits. The curves in figure A-3 will be used.

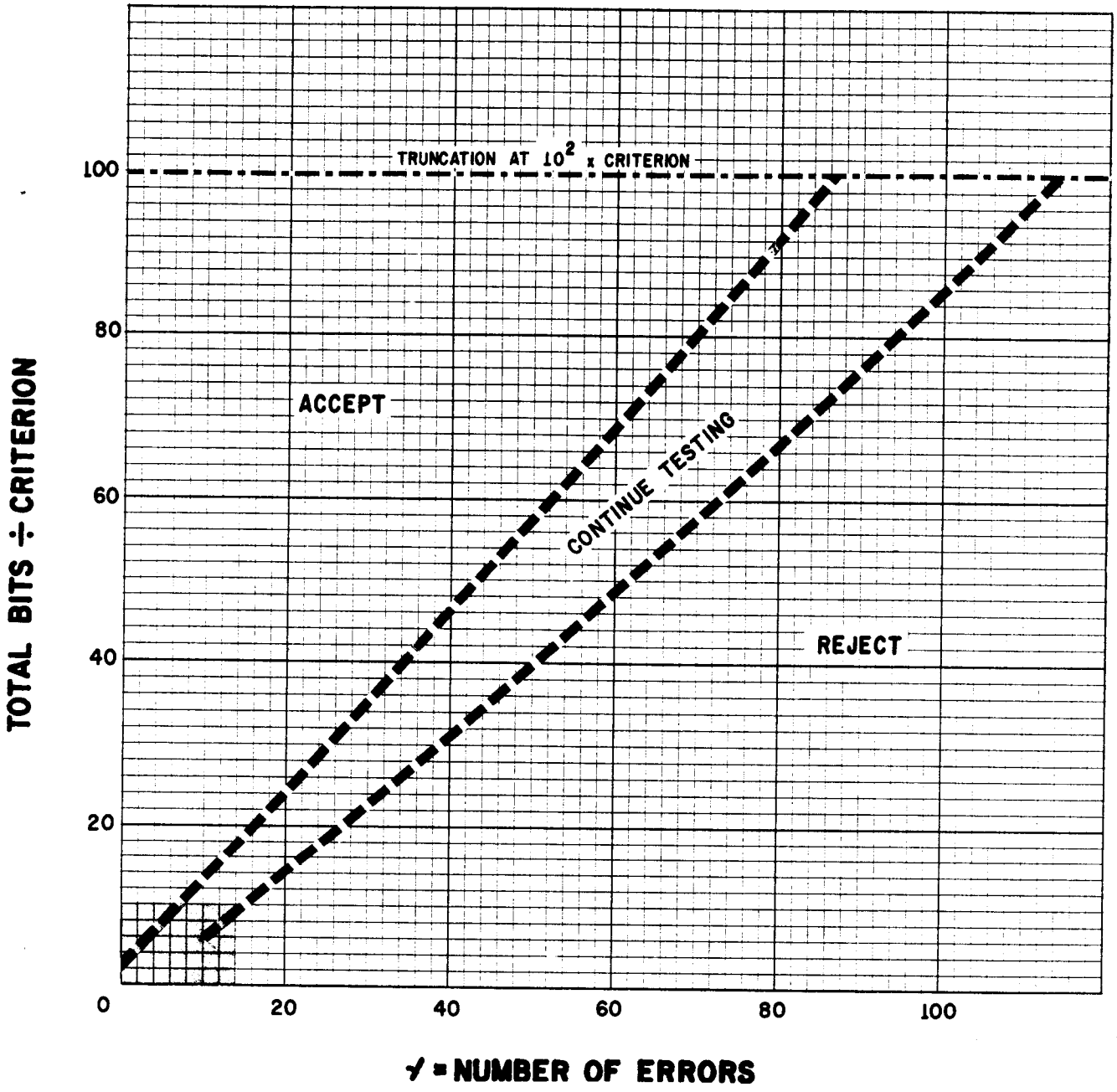
Selection of a method for any given class or type of test item will be governed principally by the following factors:

a. Specified MBBE criteria.

b. Data (bit) rate.

c. Transmission media characteristics such as radio propagation variables versus time.

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SEE TABLE A-I

Figure A-1. Data transmission test length criteria - method A.

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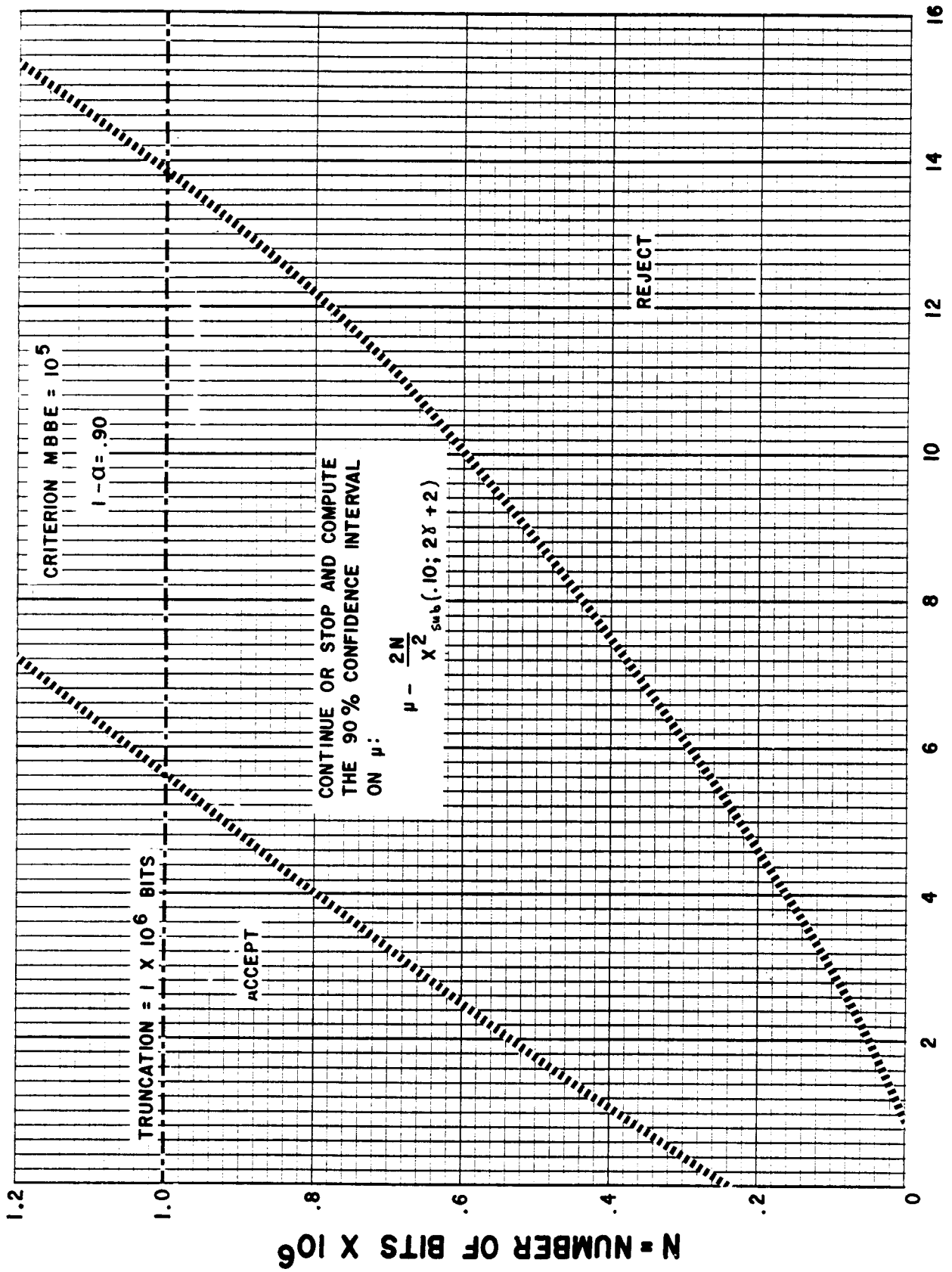


Figure A-2. Data transmission test length criteria - method B.

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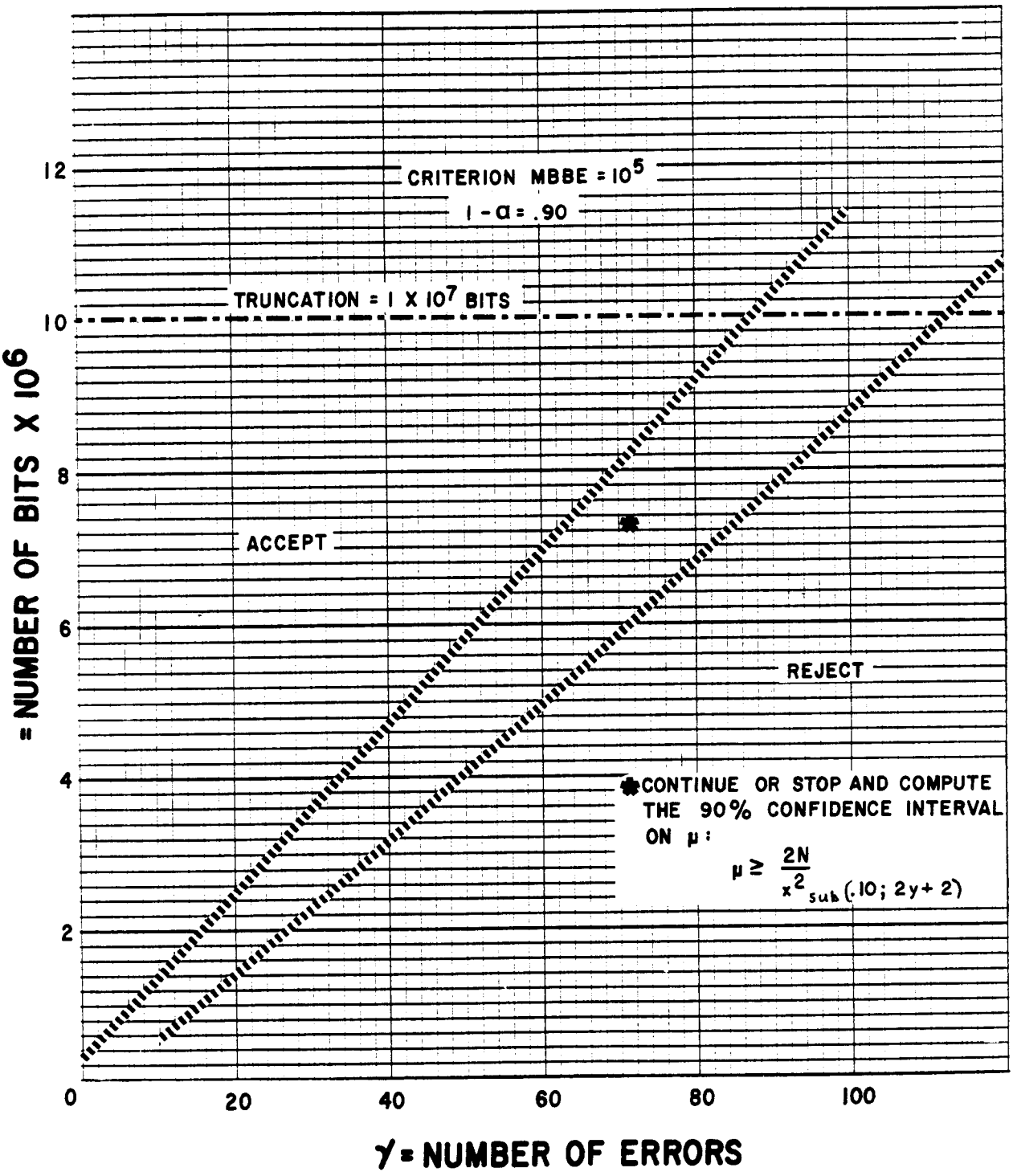


Figure A-3. Data transmission test length criteria - method C.

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APPENDIX B

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APPENDIX C

ABBREVIATIONS

ADP	Automatic Data Processing
CTP	Coordinated Test Program
I/O	Input/Output
LR	Letter Requirement
ROC	Required Operational Capability
S/N	Signal-plus-Noise-to-Noise Ratio
VF	Voice Frequency

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